

Ethical and Legal Impact

This is the ethical and legal part of the combined statement of social, ethical, legal, global and security impact.

Purpose and scope

This section assesses the ethical and legal implications of the phase-one Itqan design. It is a design review, not legal advice or a declaration of legal compliance. The assessment reflects the documented system decisions: managed external identity, an auditable consent record, in-memory recitation analysis with no retained audio, and teacher class enrolment through a join code.

The central question is not only whether the platform can identify recitation errors. It is whether it can do so without turning a learner's voice, religious practice, or school progress into a source of avoidable surveillance, embarrassment, or unfair decision-making. The system design therefore has to be evaluated together with the way it will be operated by schools, teachers, and external service providers.

1. Ethical impact

Learner wellbeing and feedback quality

Recitation feedback can affect a learner's confidence, particularly when it concerns a religious practice. The platform should present AI output as guidance rather than a judgement of ability or religious worth. It should avoid punitive automated outcomes, explain confidence where practical, and let a learner repeat an attempt or seek teacher clarification when feedback is uncertain.

A technically correct prediction can still be harmful when it is expressed without context. Feedback should distinguish a detected pronunciation pattern from a final religious ruling, use supportive language, and avoid public ranking of learners by mistakes. Where the system cannot produce a reliable result, the ethical response is to say so rather than manufacture certainty.

Agency, transparency, and teacher access

Learners need clear, Arabic-first explanations of what is collected, why it is used, and what an AI result means. Teacher dashboards should expose only the information needed for teaching. A class join code is an enrolment mechanism, not proof of informed consent or guardian authorisation; code sharing should be limited and revocable.

The platform should not make a learner dependent on an opaque score. A learner needs a route to understand an error, retry the exercise, and ask a teacher for help. Teachers should be able to treat the result as one learning signal among others, rather than as an automatic measure of competence or effort.

Children and power imbalance

Where users are minors, the school or deploying organisation must establish a verified guardian-consent and safeguarding process before collecting evaluation data or enabling teacher access. That requirement is intentionally not represented as already solved by the current data model.

Children may feel unable to refuse collection when an activity is introduced through a school or teacher. Consent material must therefore be understandable, separate from ordinary classroom access where required, and supported by a clear escalation path. A join code should not be distributed as if it were a substitute for a safeguarding decision.

Stewardship of Qur’anic content

The Qur’anic text must be presented accurately and respectfully. Content provenance, source attribution, and review of display or annotation changes are essential. Automated feedback should be framed as technical learning assistance, with qualified human instruction remaining available for material disputes.

This also means avoiding product language that treats sacred content merely as training data. Any correction rules, transliterations, or visual highlights should be reviewed for accuracy and cultural appropriateness. The team should preserve a clear link from displayed material back to its authorised source and record which version is in use.

2. Legal and data-protection impact

Personal data and consent

Saudi Arabia’s Personal Data Protection Law (PDPL) gives individuals rights to be informed, access their personal data, request correction, request destruction in applicable cases, and withdraw consent. Itqan’s **CONSENT_RECORD** should retain the purpose, version, method, grant/withdrawal status, and timestamp required to demonstrate the user’s choice. Privacy notices and product workflows must make those rights actionable, not merely documented.

For Itqan, personal data can include account identifiers, class membership, progress records, error history, and technical information associated with a learner. The legal analysis must look beyond the database schema: a privacy notice, support process, analytics configuration, and staff access practice can each determine whether the right level of protection is achieved in practice.

Data minimisation and retention

The current architecture reduces sensitivity by using managed identity (**identity_subject**) rather than locally stored password hashes and by discarding recitation audio after in-memory analysis. The operational policy must match the design: no audio recordings, raw voice samples, or debug logs containing them should be retained. Retention periods for accounts, consent records, progress data, and backups still need approval and implementation.

Data minimisation should also guide future features. A request to keep recordings for teacher review, train an AI model, or create a public showcase would change the nature of the data being handled and require a new assessment before implementation. It should not be treated as a small extension of the current no-retention design.

External processing and transfers

If an external AI analysis provider processes personal data outside Saudi Arabia, deployment must not begin until the organisation has completed a PDPL transfer assessment and implemented the required safeguard or lawful mechanism. Data-processing agreements, the provider’s retention terms, and the location of processing must be reviewed before release.

The team must know whether the provider receives only a temporary stream or also metadata, whether it uses customer inputs for model improvement, and how deletion is verified. The same review applies to managed

identity, error monitoring, analytics, and backup services. A vendor's claim of security does not remove the deploying organisation's responsibility to understand the processing arrangement.

Content and model licensing

Before public release, the team must confirm permission, attribution, and applicable terms for Qur'anic source material, reciter references, AI models, datasets, and third-party services. Referencing a source does not by itself establish redistribution rights.

Model and dataset licences need particular care because an academic prototype can move into a public product quickly. The team must document the version, source, restrictions, and attribution requirement for every component that is bundled, hosted, or invoked through an API. If a licence is unclear, the component should not be used until the uncertainty is resolved.

3. Operational accountability

Access, governance, and review

Ethical safeguards work only when somebody is responsible for operating them. The deploying organisation should define who can view class progress, who can respond to a learner or guardian request, who approves a new external provider, and who can change retention settings. Access should follow the least-privilege principle: a teacher needs the information required for teaching, while administrators need only the information required for support and governance.

This is particularly important where a school uses shared devices or a teacher has multiple classes. The product should not expose one class to another, and an old teacher relationship should not continue to give access after the class ends. Periodic access review is a practical legal and ethical control, not merely an administrative task.

Incident response and honest communication

A no-audio-retention design reduces risk but does not remove it. Account data, consent history, progress records, and vendor configuration can still be exposed or misused. Before deployment, the team needs a documented incident route: how a concern is reported, who assesses it, how affected people are informed when required, and how the cause is corrected. The route should be tested with a simple scenario rather than written and forgotten.

The platform should also communicate limitations honestly. If the AI analysis is unavailable, uncertain, or unsuitable for a particular recitation pattern, the learner should receive a clear message and a safe fallback. Hiding uncertainty can create both an ethical harm and an inaccurate record of learning progress.

Fairness, accessibility, and evaluation

The team should evaluate whether feedback quality changes across age groups, reading levels, devices, network conditions, and recognised recitation styles. A model that works well for one group but repeatedly flags another group unfairly can amplify disadvantage even if it has a good overall accuracy score. Evaluation needs representative, lawfully sourced data and review by knowledgeable educators.

Accessibility is part of the same responsibility. Instructions, consent material, error explanations, and teacher-facing controls should remain understandable for users with different levels of digital literacy. A learner who cannot understand why a result was produced cannot meaningfully challenge or learn from it.

Change control

Privacy and ethical impact should be reassessed when the purpose or data flow changes. Retaining recordings, adding biometric features, training a model on learner interactions, sharing analytics with a partner, or changing from join-code enrolment to another flow are material changes. Each must be reviewed before release, with the relevant notices, consent process, retention rule, and architecture documentation updated together.

This discipline prevents an initially narrow prototype from becoming a broader surveillance system by accident. It also gives the team a clear decision point: a feature that cannot be justified, protected, and explained should not be deployed merely because it is technically possible.

4. Design controls and remaining obligations

Area	Current design control	Required before deployment
Identity	Managed identity provider; no local password hash.	Confirm provider contract, access controls, and privacy notice.
Consent	CONSENT_RECORD captures purpose, version, method, status, and withdrawal.	Implement consent UI, withdrawal workflow, and rights-request handling.
Recitation data	Audio is analysed in memory and not stored.	Verify logs, monitoring, backups, and vendors do not retain audio.
Classes	Join code supports enrolment without invitation links.	Protect code disclosure; establish verified guardian/school process for minors.
AI service	External analysis may be used for feedback.	Complete transfer, vendor, accuracy, and human-escalation review.
Qur'anic content	Source provenance is documented.	Validate attribution, terms, and release permissions.

5. Conclusion

Itqan's phase-one choices reduce several major risks: it avoids local passwords, keeps a consent history, and does not retain recitation audio. These controls are not sufficient on their own for deployment. The open obligations above—especially minors' safeguarding, operational deletion controls, vendor/transfer review, and content licensing—must be completed before handling real learner data.

References

Saudi Data & AI Authority (SDAIA), *Personal Data Protection Law and Regulations*: <https://sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx>

SDAIA, *Data Subject Rights under the PDPL*: <https://dgp.sdaia.gov.sa/wps/portal/pdp/knowledgecenter/details/PDPLCP/>

SDAIA, *Standard Contractual Clauses for Personal Data Transfer*: <https://sdaia.gov.sa/Documents/StandardContractualClausesForPersonalDataTransferEN.pdf>

King Fahd Glorious Qur'an Printing Complex, *Vision*: <https://qurancomplex.gov.sa/en/kfgqpc/vision/>

Prepared for the Itqan graduation project. Legal applicability and compliance decisions require review by the deploying organisation and qualified legal counsel.

Security Impact — Design Correction

This correction supersedes the listed passages in the earlier Security Impact section.

Final architecture decisions

The class-enrolment and consent wording in the original security report pre-dated the final ERD and class-diagram decisions. This page makes the final system boundary explicit. All other security positions—including managed identity, no retained recitation audio, access authorisation, and the need for deployment controls—remain unchanged.

Earlier wording	Final phase-one design
A learner or guardian accepts an invitation.	A learner joins a class using its join code. There are no invitation links in the phase-one model.
The system collects age to determine guardian consent and verifies the guardian.	No age or guardian relationship is represented in the current model. A deploying school must establish verified guardian consent and safeguarding before enabling minor access or evaluation.
Teacher access rests on guardian acceptance.	Teacher access rests on the learner’s active class enrolment and role-based authorisation. A join code is not evidence of consent.
The analysis service runs outside Saudi Arabia.	External processing remains a deployment decision. It must not begin until the transfer and vendor review required by the PDPL has been completed.

Use in the combined report

Read this page as part of the Security Impact section. It resolves the contradictions without claiming that guardian verification, cross-border processing, or vendor contracts are already implemented in the phase-one prototype.

Security Impact

This is the security part of the combined statement of social, ethical, legal, global and security impact.

1. What Makes Itqan a Security Question

Most of what Itqan holds is unremarkable: names, progress figures, which surahs a learner has practised. One thing is not. To do its work, the system listens to a person reciting the Quran.

That places it in an unusual position. A recording of recitation is a recording of a voice, which identifies the person who made it. It is also a record of religious practice, and specifically a record of someone practising imperfectly — making mistakes they are trying to correct. And a substantial share of Itqan's users are expected to be children, since Juz Amma is what beginners learn first.

The security question for this project is therefore not principally about servers and passwords. It is about what the system chooses to keep.

2. The Decision That Shapes Everything Else

Itqan does not store recitation audio. Audio is held in memory only for as long as the analysis takes, and is then discarded. It is never written to disk, never written to the database, never included in logs or error reports, and never made available to a teacher.

What persists is the result of the analysis: which word contained an error, what kind of error it was, and the resulting score. That is what the product actually needs — it drives the mastery percentage, the error history and the teacher's view — and it is not a voice recording.

This single decision removes most of the system's security exposure:

Risk that would exist	Status
A breach exposing recordings of identifiable people	Cannot occur; no recordings exist
Recordings of children being disclosed	Cannot occur
Recordings being retained longer than promised	Cannot occur; nothing is retained
Recordings being reused for a purpose the user did not agree to	Cannot occur
A user unable to get their recordings deleted	Cannot occur

The system also does not compute voiceprints, speaker embeddings, or any other feature that would identify a speaker from their voice. Nothing Itqan does requires it, and introducing it would change the nature of the data being handled.

This is a design position, not a limitation. It was chosen deliberately, and it is stronger than protecting stored recordings well, because it removes the asset rather than defending it.

3. Threats and How the Design Answers Them

3.1 A teacher seeing a student who is not theirs

This is the most likely serious failure in a system of this shape, and it is a failure of authorisation rather than of encryption.

The risk: a teacher requests a student's results directly — by guessing or altering an identifier — and receives data for a learner they have no relationship with.

The design response: a teacher's access depends on the combination of teacher, student, class and an **active enrolment period**, and that combination is checked on every read, in the data layer. It is not a restriction applied in the interface, which could be bypassed by requesting the data another way. Enrolment begins when the student — or a guardian, for a child — accepts an invitation, and ends when the enrolment ends. Identifiers are not sequential, so they cannot be enumerated.

Two automated tests hold this in place: a teacher cannot read a student who is not theirs, and a former teacher cannot read a former student.

3.2 Account takeover

The risk: someone gains access to a learner's account and sees their practice history, or a teacher's account and sees a whole class.

The design response: authentication is delegated to a managed identity provider rather than written by the project. Password storage, reset flows, and session handling are where systems of this size most often fail, and a managed provider removes that surface. It also supports proper session revocation, which matters for a reason beyond security: withdrawal of consent has to actually take effect, not wait for a session to expire on its own.

One scenario is treated as ordinary rather than exceptional: a children's Quran application will be used on shared family devices. Sessions therefore expire after a period of inactivity rather than persisting indefinitely, and an account-level change requires re-authentication.

3.3 Abuse of the analysis service

The risk: the analysis service is the only expensive part of the system. Someone sending large volumes of audio to it could exhaust the project's compute budget — and, because the service is rented per second, that is a financial risk rather than merely a technical one.

The design response: requests to the analysis service are authenticated and attributed to a user, rate-limited per user, and subject to a daily ceiling. A billing alert provides notice before a limit is reached rather than after.

3.4 Audio retention creeping back in

The risk: a system that is designed not to retain audio can begin retaining it by accident — a temporary file that is not cleaned up after a crash, a log that captures a request body, a backup that preserves something the live system deleted.

The design response: deletion is structural rather than scheduled. Audio exists only within the scope of a single request and is released when that request ends, including when it fails. Request bodies are excluded from logging and from error reporting.

3.5 Automated judgement about a person

The risk: the mastery percentage is a number a machine assigns to a person's religious practice. Treated carelessly, it could become a gate — a pass or fail, a consequence applied automatically.

The design response: the mastery percentage is feedback to a human being, presented to the learner and to their teacher. It does not trigger any automatic consequence. This matters particularly because many users are children, and because the system's accuracy is measured rather than perfect.

4. Legal and Regulatory Context

Itqan processes personal data in Saudi Arabia, and the Personal Data Protection Law applies. The obligations most relevant to this system are consent for a stated purpose, limiting collection to what the purpose requires, stating how long data is kept, and allowing a person to see and delete their data.

The architecture addresses these as follows:

- **Consent** is recorded separately for each purpose, with the time and the method it was given. Consent to have recitation analysed is separate from consent for a teacher to see the results.
- **Collection is limited** to what a stated purpose requires. The system does not collect age beyond what is needed to determine whether a guardian must consent, and does not collect location, contacts or advertising identifiers.
- **Retention** is stated for every kind of data the system holds. For recitation audio the answer is that it is not retained at all.
- **Deletion** is available to every user at no cost. This is worth stating explicitly, because it is not universal: at least one established application in this category places both deletion of recordings and opting out of AI training behind a paid subscription.

Children. Where a user is a minor, consent is given by a verified guardian rather than by the child, and a teacher's access to a child's results rests on the guardian's acceptance. Children's data attracts stricter treatment under Saudi rules and under the mobile application stores' own policies, and the project's approach is to reduce what is collected rather than to build additional protections around a larger collection.

Data leaving the Kingdom. The analysis service runs on a rented service outside Saudi Arabia, so recitation audio travels outside the Kingdom to be processed, although it is not stored there or anywhere else. Transfers of personal data abroad are regulated, and the formal mechanisms available are not practically obtainable by an undergraduate project team. Two things limit the exposure: nothing is retained at either end, and the audio is not accompanied by identifying information beyond what the request requires. Hosting the analysis service on departmental hardware inside the Kingdom would remove the question entirely and remains the stronger arrangement; it was not adopted because it depends on hardware the project does not control.

Classification of the audio. Whether a recording of Quranic recitation constitutes ordinary personal data or falls into a more sensitive category is genuinely arguable: it is a voice, which identifies its speaker, and it is a record of religious practice. Rather than rest on the more favourable reading, Itqan is designed as though the stricter one applies. Not keeping the audio makes the question moot in practice, which is the safer position to be in.

5. Impact Beyond the System

The handbook asks for the impact of the project on society, not only on its users.

On learners. The most direct impact is that a person can practise recitation, be told where they went wrong, and not have a recording of that mistake exist afterwards. For an adult this is a matter of dignity; for a child it also removes a record that could follow them.

On the category. Applications that listen to Quran recitation are becoming common. The prevailing pattern is to retain recordings — often to improve the vendor's own models, sometimes with the ability to opt out or delete reserved for paying users. Itqan demonstrates that the corrective function does not require retention: the analysis can be performed and the recording discarded. As a graduation project it changes nothing by itself, but it is evidence that the trade-off usually presented as necessary is not.

On the teacher–learner relationship. Giving a teacher visibility into a learner's practice is useful and is what the Al-Mu'allim mode exists for. It is also a form of supervision, and supervision without limits becomes surveillance. The design draws the line deliberately: a teacher sees error types, word locations and progress, for their own students, for the period they are enrolled. A teacher does not hear the student, and does not see activity outside that relationship. The learner or guardian can end the relationship, and the access ends with it.

On trust in the religious context. A system that tells people they have recited the Quran incorrectly carries a responsibility to be careful. Itqan's accuracy is measured and reported rather than asserted, its tajweed feedback is limited to rules for which reliability has been established, and it is designed to stay silent rather than offer a correction it is not confident in. A false correction in this context is more damaging than no correction.

6. Residual Risks

The following remain, and are stated rather than concealed.

Risk	Why it remains	What reduces it
Recitation audio transits outside the Kingdom	The analysis service is hosted on rented infrastructure abroad	Nothing is retained at either end; moving the service in-Kingdom would remove it
Account identifiers are held by a managed identity provider abroad	Consequence of not writing the login system	Identifiers only; no recitation data is held there
The system can flag an error the user did not make	No speech system is perfect; the model's own documentation acknowledges specific weaknesses	Accuracy is measured and published; the system prefers silence to a low-confidence correction
Security depends on correct implementation, not only on design	The access rule must be applied at every endpoint, not merely designed	Automated tests for the access rule; the rule is enforced in one place rather than repeated
A guardian's consent is only as good as the verification behind it	Verifying that an adult is genuinely a guardian is unsolved generally	Consent is recorded with its method, so the basis of any access is auditable

7. Summary

Itqan's security position rests on one decision taken early: the system does not keep the recordings it analyses. That decision removes the asset that would otherwise have been the most sensitive thing the project holds, and with it most of the obligations and most of the consequences of failure.

What remains is a conventional set of concerns — authorisation, authentication, abuse of an expensive service — addressed in conventional ways, with one exception worth naming. Teacher access is scoped to an accepted, time-bounded relationship and enforced on every read, because the difference between supporting a learner and monitoring them is exactly the difference between access that is granted and access that is assumed.

Social and Global Impact

This is the social and global part of the combined statement of social, ethical, legal, global and security impact.

Purpose and scope

This section considers how phase-one Itqan may affect learners, teachers, families, and wider communities. Itqan is an AI-assisted Qur'an recitation learning platform: it provides guided practice, feedback, progress information, teacher-class support, and source-aware Qur'anic content. Its impact depends as much on access, language, and classroom use as on technical accuracy.

The analysis avoids claiming that software alone solves an educational or social need. The platform can support practice and extend access to structured feedback, but it cannot replace a qualified teacher, a supportive family environment, or an inclusive learning setting. Success means improving learning opportunities without creating a new barrier for learners who have less connectivity, confidence, or digital access.

1. Social impact

Supporting practice between lessons

Itqan can give learners an opportunity to practise recitation outside a scheduled lesson and receive immediate, structured feedback. That may be particularly useful where a teacher has limited time with each learner or where a learner is reluctant to make mistakes in front of a group. Progress history can help a learner see improvement over time rather than treating one imperfect attempt as failure.

The social value comes from practice being more available, not from replacing human teaching. Teacher review, encouragement, and context remain essential. The product should therefore describe itself as a learning aid and make it easy for a teacher to interpret results alongside observation, not as a system that assigns a final ability label.

Language, culture, and belonging

Arabic-first interaction is important because the platform concerns Qur'anic learning and is intended for users who may prefer Arabic explanations. Labels, feedback, help content, and consent material should be understandable in the learner's language rather than relying on a translated technical interface. Clear wording also reduces the risk that an error message is experienced as criticism of the learner or of their religious identity.

Respectful presentation of Qur'anic material contributes to cultural continuity. Source attribution and accurate text presentation allow learners to connect the digital experience to trusted scholarship rather than to an anonymous automated system. The product must not present a single technical output as more authoritative than qualified instruction or established recitation practice.

Teacher and family relationships

Class features can support teachers by showing patterns that may deserve attention, such as a repeated difficulty with an exercise. They should not create a pressure to monitor every learner continuously. A useful dashboard prioritises teaching actions, minimises unnecessary comparison, and avoids public leaderboards that can shame a learner who is progressing more slowly.

Family involvement can be positive when it encourages practice, but the product should not assume every learner has equal support at home. Where minors use the platform, schools and families need an agreed safeguarding and communication process. A class join code enables enrolment; it does not prove that a guardian has understood the data use or agreed to it.

Inclusion and the digital divide

A digital learning tool can widen inequality if it assumes that every household has a recent device, reliable connectivity, private space to practise, or a digitally confident adult. The platform should remain usable on common devices, avoid unnecessary bandwidth use, and provide a clear fallback when analysis cannot run. Schools should avoid making home access a hidden requirement for participation or assessment.

Inclusion also includes learners with different reading needs, levels of Arabic literacy, and confidence with technology. Short instructions, meaningful error explanations, consistent controls, and teacher-supported alternatives make the benefit more evenly available. Usage data should never be used to infer that a learner is less committed simply because they have fewer opportunities to connect.

2. Global impact

Contribution to inclusive learning

The United Nations Sustainable Development Goal 4 calls for inclusive and equitable quality education and recognises that access gaps connected to location, income, disability, and connectivity remain significant. Itqan can contribute in a limited but practical way by making structured practice material and feedback more portable. It should make that contribution without claiming to solve wider educational inequality.

A global-ready product is not simply one that can be opened in many countries. It must be understandable across different educational settings, account for unequal access to devices and networks, and avoid assuming that one classroom practice is universal. Deployment beyond the original context should be gradual, with local educator input rather than a one-size-fits-all rollout.

Cultural reach and local adaptation

Qur'anic learning connects communities across countries, languages, and educational traditions. That reach creates an opportunity to support learners who do not have easy access to structured practice, but it also creates a responsibility to avoid flattening cultural differences. Examples, terminology, explanatory content, and feedback language should be reviewed with educators familiar with the intended community.

The system should be designed for adaptation without silently changing the underlying content source or feedback rules. A localised interface must preserve provenance and make clear what has changed: language, examples, teaching guidance, or the technical model. Where the platform cannot support a local requirement reliably, it should say so rather than imply universal suitability.

Environmental and operational footprint

The platform's direct environmental impact is modest at prototype scale, but AI analysis, cloud hosting, and repeated media processing still consume energy. The existing decision not to store recitation audio reduces storage demand and unnecessary duplication. Efficient processing, sensible retry limits, and avoiding

collection that is not needed for learning are practical ways to keep the system proportionate as it grows.

Global operation also creates dependency on external platforms. Managed identity and external analysis can improve reliability, but they can create outages, cost changes, and cross-border data concerns. The product should retain clear ownership of its core learning records, document each dependency, and communicate service limitations rather than hiding them from schools or learners.

3. Responsible deployment

Evidence before broad claims

Before promoting Itqan as improving outcomes, the team should test it with representative learners and educators. Evaluation should look beyond technical error detection to questions such as: Does it help a learner practise with more confidence? Can teachers understand the feedback? Does it create a disadvantage for learners with poorer devices or networks? The answers should guide improvement and should not be inferred from usage counts alone.

Any research activity involving children, classroom data, or recorded interactions needs its own ethics and consent process. Academic evaluation is not automatically covered by ordinary product enrolment. The team should publish only aggregated, non-identifying findings and should avoid presenting early prototype results as evidence of broad social benefit.

Community accountability

Feedback from learners, teachers, parents, and qualified Qur'an educators should be treated as part of the product's quality process. A simple route to report confusing feedback, cultural concerns, or accessibility barriers gives affected people a way to influence the system. Reports should be reviewed, categorised, and linked to a decision rather than collected without response.

The strongest global posture is humility: deploy where the team can provide support, listen to local feedback, and correct problems. Expansion should pause if the platform cannot safeguard learners, explain its data practices, or obtain appropriate educational and cultural review in the new context.

4. Design controls and remaining obligations

Area	Potential contribution	Required before deployment
Learning practice	Guided practice and feedback between lessons.	Keep teacher review and a learner retry/escalation path.
Language and culture	Arabic-first, source-aware learning experience.	Review wording and content with qualified educators.
Access	Portable support on common devices.	Test low-bandwidth and school-supported alternatives.
Classes	Teacher insight into practice patterns.	Avoid ranking; protect code sharing and minor safeguarding.
Global use	Potential access across communities.	Use local review and validate language, data, and support needs.
Evaluation	Evidence for iterative improvement.	Obtain separate ethics approval and publish aggregate findings only.

5. Conclusion

Itqan can support more flexible, respectful Qur'an recitation practice, but the benefit is conditional. It must remain accessible, Arabic-first, teacher-supported, culturally accountable, and honest about its limitations. The design choices that reduce data collection and avoid automatic judgement are also social choices: they help keep the platform focused on learning rather than surveillance or ranking.

References

United Nations, *Sustainable Development Goal 4: Quality Education*: <https://sdgs.un.org/goals/goal4>

UNESCO Global Education Monitoring Report, *Technology in Education*: <https://www.unesco.org/gem-report/en/publication/technology>

King Fahd Glorious Qur'an Printing Complex, *Vision*: <https://qurancomplex.gov.sa/en/kfgqpc/vision/>

Prepared for the Itqan graduation project. This assessment describes intended impacts and deployment conditions; it does not claim measured educational outcomes.